

Aggregate Management:

Listing the aggregate:

```
10.1.1.130 - PuTTY
prodcluster:>
prodcluster:> storage aggregate show
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0	3.34GB	1.54GB	54%	online	1	prodcluster-01	raid_dp, normal
aggr1	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal
dept2aggr	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal

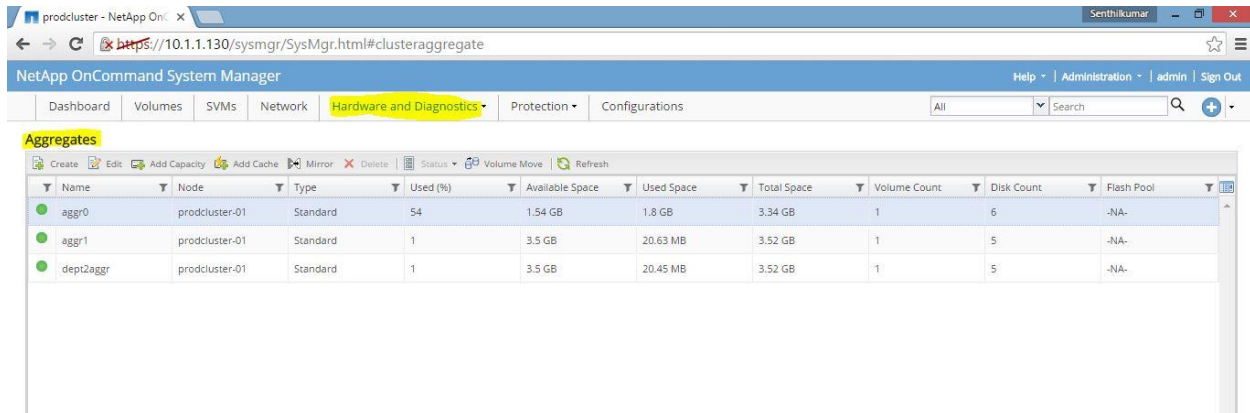
```
3 entries were displayed.
```

Listing the particular aggregate information:

```
10.1.1.130 - PuTTY
prodcluster:>
prodcluster:>
prodcluster:> storage aggregate show -aggregate dept2aggr
```

```
Aggregate: dept2aggr
Storage Type: hdd
Checksum Style: block
Number Of Disks: 5
Mirror: false
Disks for First Plex: NET-1.19, NET-1.27,
NET-1.20, NET-1.28,
NET-1.21
Disks for Mirrored Plex: -
Partitions for First Plex: -
Partitions for Mirrored Plex: -
Node: prodcluster-01
Free Space Reallocation: off
HA Policy: sfo
Ignore Inconsistent: off
Space Reserved for Snapshot Copies: -
Aggregate Nearly Full Threshold Percent: 95%
Aggregate Full Threshold Percent: 98%
Checksum Verification: on
```

List aggregate information using System Manager.



Name	Node	Type	Used (%)	Available Space	Used Space	Total Space	Volume Count	Disk Count	Flash Pool
aggr0	prodcluster-01	Standard	54	1.54 GB	1.8 GB	3.34 GB	1	6	-NA-
aggr1	prodcluster-01	Standard	1	3.5 GB	20.63 MB	3.52 GB	1	5	-NA-
dept2aggr	prodcluster-01	Standard	1	3.5 GB	20.45 MB	3.52 GB	1	5	-NA-

Creating an Aggregate using disk type and raidtype.

```
prodcluster::> storage aggregate create -aggregate aggrtest -disktype ATA -diskcount 5 -raidtype raid4

Info: The layout for aggregate "aggrtest" on node "prodcluster-01" would be:

First Plex

RAID Group rg0, 5 disks (block checksum, raid4)
  Position  Disk              Type              Size
  -----  -
  parity    NET-1.36             ATA               -
  data      NET-1.37             ATA               1.95GB
  data      NET-1.38             ATA               1.95GB
  data      NET-1.39             ATA               1.95GB
  data      NET-1.40             ATA               1.95GB

Aggregate capacity available for volume use would be 7.03GB.

Do you want to continue? {y|n}: y
[Job 53] Job succeeded: DONE

prodcluster::>
```

```
prodcluster::> storage aggregate show
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0	3.34GB	1.54GB	54%	online	1	prodcluster-01	raid_dp, normal
aggr1	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal
aggrtest	7.03GB	7.03GB	0%	online	0	prodcluster-01	raid4, normal
dept2aggr	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal

4 entries were displayed.

```
prodcluster::>
```

Change the status of Aggregate.

```
prodcluster::> storage aggregate restrict -aggregate aggrtest
```

```
prodcluster::> storage aggregate show
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0	3.34GB	1.54GB	54%	online	1	prodcluster-01	raid_dp, normal
aggr1	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal
aggrtest	0B	0B	0%	restricted	0	prodcluster-01	raid4, normal
dept2aggr	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal

4 entries were displayed.

```
prodcluster::>
```

Aggregate offline.

```
prodcluster::> storage aggregate offline -aggregate aggrtest
Aggregate offline successful on aggregate: aggrtest
prodcluster::> storage aggregate show
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0	3.34GB	1.54GB	54%	online	1	prodcluster-01	raid_dp, normal
aggr1	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal
aggrtest	0B	0B	0%	offline	0	prodcluster-01	raid4, normal
dept2aggr	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal

4 entries were displayed.

```
prodcluster::>
```

Enabling the aggregate online.

```
prodcluster::> storage aggregate online -aggregate aggrtest
Aggregate online successful on aggregate: aggrtest
prodcluster::> storage aggregate show
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0	3.34GB	1.54GB	54%	online	1	prodcluster-01	raid_dp, normal
aggr1	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal
aggrtest	7.03GB	7.03GB	0%	online	0	prodcluster-01	raid4, normal
dept2aggr	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal

4 entries were displayed.

```
prodcluster::>
```

Viewing the aggregate space.

```
prodcluster:~>
prodcluster:~> storage aggregate show-space -aggregate aggrtest

Aggregate Name: aggrtest
Volume Footprints: 0B
Volume Footprints Percent: 0%
Total Space for Snapshot Copies in Bytes: 0B
Space Reserved for Snapshot Copies: 0%
Aggregate Metadata: 204KB
Aggregate Metadata Percent: 0%
Total Used: 204KB
Total Used Percent: 0%
Size: 7.03GB
Snapshot Reserve Unusable: -
Snapshot Reserve Unusable Percent: -
Total Physical Used Size: 304KB
Physical Used Percentage: 0%

prodcluster:~>
```

Adding the capacity to an existing Aggregate.

```
prodcluster:~>
prodcluster:~> storage aggregate add-disks -aggregate aggrtest -diskcount 1

Info: Disks would be added to aggregate "aggrtest" on node "prodcluster-01" in
the following manner:

First Plex

RAID Group rg0, 6 disks (block checksum, raid4)
Position  Disk                      Type                      Size
-----
data      NET-1.41                      ATA                      1.95GB

Aggregate capacity available for volume use would be increased by 1.76GB.

Do you want to continue? {y|n}: y

prodcluster:~>
```

```

prodcluster:~>
prodcluster:~> storage aggregate show-space -aggregate aggrtest

Aggregate Name: aggrtest
Volume Footprints: 0B
Volume Footprints Percent: 0%
Total Space for Snapshot Copies in Bytes: 0B
Space Reserved for Snapshot Copies: 0%
Aggregate Metadata: 164KB
Aggregate Metadata Percent: 0%
Total Used: 164KB
Total Used Percent: 0%
Size: 8.79GB
Snapshot Reserve Unusable: -
Snapshot Reserve Unusable Percent: -
Total Physical Used Size: 436KB
Physical Used Percentage: 0%

prodcluster:~>

```

Creating an new aggregate using System Manager.

The screenshot displays the NetApp OnCommand System Manager interface. The 'Create Aggregate' dialog is open, prompting the user to select a disk type and specify the number of disks. A 'Select Disk Type' sub-dialog is open, showing a table of available disk types:

Disk Ty...	Node	Disk Size	RPM	Checks...	Pool	Available ...	Total Capac...
FSAS	prodcluste...	1.97 GB	10000	block	Pool 0	5	9.87 GB
FCAL	prodcluste...	1020.5 MB	15000	block	Pool 0	12	11.96 GB
SSD	prodcluste...	520.5 MB	-NA-	block	Pool 0	13	6.61 GB

The main dialog also shows the aggregate name 'aggrnew' and a 'Browse' button for disk selection. The background interface shows the 'Aggregates' section with a list of existing aggregates: aggr0, aggr1, aggrtest, and dept2aggr.

Creating an Aggregate using RAID TEC Level.

Create Aggregate

To create an aggregate, select a disk type then specify the number of disks.

Change RAID Configuration

RAID Type: RAID-TEC

RAID Group Size: 7 Disks Min : 7, Max : 29 for RAID-TEC aggregate with FCAL disks.

RAID Allocation: 11 disks or partitions in 2 RAID groups.
■ Parity 5.98 GB ■ Data 4.39 GB □ Empty Slots
rg0 ■■■■■■
rg1 ■■■■■■□□□□
i "rg" indicates a RAID group

Save Cancel

Create Cancel

prodcluster - NetApp OnCommand System Manager

Help Administration admin Sign Out

Dashboard Volumes SVMs Network Hardware and Diagnostics Protection Configurations

Aggregates

Create Edit Add Capacity Add Cache Mirror Delete Status Volume Move Refresh

Name	Node	Type	Used (%)	Available Space	Used Space	Total Space	Volume Count	Disk Count	Flash Pool
aggr0	prodcluster-01	Standard	54	1.54 GB	1.8 GB	3.34 GB	1	6	-NA-
aggr1	prodcluster-01	Standard	1	3.5 GB	20.46 MB	3.52 GB	1	5	-NA-
aggrnew	prodcluster-01	Standard	0	4.39 GB	148 KB	4.39 GB	0	11	-NA-
aggrtest	prodcluster-01	Standard	0	8.79 GB	164 KB	8.79 GB	0	6	-NA-
dept2aggr	prodcluster-01	Standard	1	3.5 GB	20.45 MB	3.52 GB	1	5	-NA-

Name: aggrnew
 Status: online
 RAID Type: RAID-TEC, normal
 Root: No
 Files: 96
 Maximum Files: 31136
 Checksum: block

Details Volumes Disk Layout Performance

```
prodcluster::>
prodcluster::> storage aggregate show
```

Aggregate	Size	Available	Used%	State	#Vols	Nodes	RAID Status
aggr0	3.34GB	1.54GB	54%	online	1	prodcluster-01	raid_dp, normal
aggr1	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal
aggrnew	4.39GB	4.39GB	0%	online	0	prodcluster-01	raid_tec, normal
aggrtest	8.79GB	8.79GB	0%	online	0	prodcluster-01	raid4, normal
dept2aggr	3.52GB	3.50GB	1%	online	1	prodcluster-01	raid4, normal

5 entries were displayed.

```
prodcluster::>
```

Listing the Aggregate disk information.

```
10.1.1.130 - PuTTY
prodcluster::> storage aggregate show-status -aggregate aggrnew

Owner Node: prodcluster-01
Aggregate: aggrnew (online, raid_tec) (block checksums)
Plex: /aggrnew/plex0 (online, normal, active, pool0)
RAID Group /aggrnew/plex0/rg0 (normal, block checksums)

      Position Disk                Pool Type      RPM      Usable Physical
atus -----
-----
      tparity NET-1.22                0  FCAL      15000    1020MB    1.00GB (n
ormal)
      dparity NET-1.29                0  FCAL      15000    1020MB    1.00GB (n
ormal)
      parity  NET-1.23                0  FCAL      15000    1020MB    1.00GB (n
ormal)
      data    NET-1.31                0  FCAL      15000    1020MB    1.00GB (n
ormal)
      data    NET-1.24                0  FCAL      15000    1020MB    1.00GB (n
ormal)
      data    NET-1.32                0  FCAL      15000    1020MB    1.00GB (n
ormal)
      data    NET-1.25                0  FCAL      15000    1020MB    1.00GB (n
```

Modifying an existing aggregate RAID type.

```
prodcluster::> aggr modify -aggregate aggrtest -raidtype raid_dp

prodcluster::> storage aggregate show

Aggregate      Size Available Used% State  #Vols  Nodes      RAID Status
-----
aggr0          3.34GB    1.54GB    54% online    1 prodcluster-01  raid_dp,
normal
aggr1          3.52GB    3.50GB     1% online    1 prodcluster-01  raid4,
normal
aggrnew        4.39GB    4.39GB     0% online    0 prodcluster-01  raid_tec,
normal
aggrtest       8.79GB    8.79GB     0% online    0 prodcluster-01  raid_dp,
reconstruct
dept2aggr      3.52GB    3.50GB     1% online    1 prodcluster-01  raid4,
normal

5 entries were displayed.

prodcluster::>
```

Deleting an existing Aggregate.

```
prodcluster::>
prodcluster::> storage aggregate delete -aggregate aggrnew

Warning: Are you sure you want to destroy aggregate "aggrnew"?
{y|n}: y
[Job 55] Job succeeded: DONE

prodcluster::> storage aggr show

Aggregate      Size Available Used% State  #Vols  Nodes      RAID Status
-----
aggr0          3.34GB    1.54GB   54% online    1 prodcluster-01  raid_dp,
normal
aggr1          3.52GB    3.50GB    1% online    1 prodcluster-01  raid4,
normal
aggrtest       8.79GB    8.79GB    0% online    0 prodcluster-01  raid_dp,
normal
dept2aggr      3.52GB    3.50GB    1% online    1 prodcluster-01  raid4,
normal
4 entries were displayed.

prodcluster::>
```

Disk Type	Group Type	Default	Maximum
SATA*	RAID4	7	7
	RAID-DP	14	20
	RAID-TEC	21	29
NL-SAS*	RAID4	7	7
	RAID-DP	14	20
	RAID-TEC	21	29
SAS	RAID4	8	14
	RAID-DP	16	28
	RAID-TEC	24	29
SSD	RAID4	8	14
	RAID-DP	23	28
	RAID-TEC	24	29